SAVEETHA
INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

Answer Script Evaluation Sheet

Continuous Internal Assessment 1

Date: 5.11.26

| Programme | B.E/ B. Tech | Slot | C |
|---|---|---|---|
| Course Code | CSA0709 | Course Name | Computer Networks |
| Duration | 1.30 Hrs | Marks | 50 |

| Q. No. | Max. Marks | Awarded Marks |
|---|---|---|
| 1. | 10 | 10 |
| 2. | 10 | 10 |
| 3. | 10 | 9 |
| 4. | 10 | 9 |
| 5. | 10 | 9 |
| Total | 50 | 46 |

**Course Faculty:** Dr. Shankar R.,

SAVEETHA
INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

Programme: **B.E / B.Tech**

Course Code: **CSA07**

Course Name: **Computer Networks**

Duration: **1: 30 Hr**

Date : 03.11.2026 Max Marks: **50 Marks**

**CONTINUOUS INTERNAL ASSESSMENT 1**

**Course Outcomes**

On successful completion of the course, students will be able to:

| CO | Course Outcome Statement |
|---|---|
| CO1 | Explain the layered network architectures (OSI/TCP-IP), the functions of each layer, and the fundamentals of data communication in computer networks. |
| CO2 | Apply IP addressing, subnetting, and routing concepts to design and configure interconnected networks. |

**Answer All Questions**

1. Answer the following questions on the OSI model and Transport layer services.

**(a)** Explain the primary functions of the Data Link layer (framing, physical (MAC) addressing, medium access control, error detection using CRC, and hop-to-hopflow control) and of the Transport layer (process-toprocess delivery using port numbers, segmentation and reassembly, end-to-end flow control, congestion control, and error recovery through acknowledgements and retransmission). Analyze how the two layers work together to give error-free and reliable data delivery: the Data Link layer detects and discards corrupted frames on each individual link, while the Transport layer provides end-to-end reliability across multiple links and routers by using sequence numbers, ACKs, and retransmission of lost segments. Use a layered diagram to support the explanation? *(6 Marks)*

**(b)** Differentiate between connection-oriented service (TCP: handshake before transfer, reliable, ordered, flow and congestion control) and connectionless service (UDP: no setup, best-effort, low overhead) at the Transport layer. Compare them on connection setup, reliability, speed, header size, and state maintained. Give real-world examples: TCP for web browsing (HTTP/HTTPS), email (SMTP), and file transfer (FTP), and UDP for DNS queries, VoIP, video streaming, and online gaming. Present the comparison in a tab le. *(4 Marks)*

***(10) (CO 1) (BL2) (BL4)***

2. A data packet travels from a source host to a destination host across multiple routers.

**(a)** Describe the concept of encapsulation, in which each OSI layer at the source adds its own header (and a trailer at the Data Link layer) to the data received from the layer above, producing the PDUs: data, segment or datagram, packet, frame, and bits. Describe decapsulation, in which each layer at the destination strips its corresponding header and passes the payload upward until the application receives the original data. Explain what happens at intermediate routers (the frame is decapsulated up to the Network layer, the next hop is determined, and a new frame is built). Support the answer with a neat diagram showing the headers at each layer. *(6 Marks)*

**(b)** Identify the Network layer as the layer responsible for routing packets across interconnected networks. Explain how it uses logical addressing (IP addresses) to identify the source and destination networks and hosts, how routers use routing tables and the destination IP address to select the best path, and how logical (IP) addresses differ from physical (MAC) addresses, which change at each hop while IP addresses stay the same end to end. Mention fragmentation and the TTL field briefly. *(4 Marks)*

***(10) (CO1) (BL2)***

3. Answer the following questions on IP addressing and subnetting.

**(a)**Routers and Layer 3 devices between a source and a destination system forward packets using IP addresses and routing tables. Explain the limitations of classful IP addressing: fixed network and host boundaries (Class A, B, C), wasted addresses (a Class C block of 254 hosts is too small and a Class B block of 65,534 is too large for most organizations), rapid IPv4 address exhaustion, inflexible subnetting, and the growth of routing tables on routers. Explain how Classless Inter-Domain Routing (CIDR) overcomes these problems through variable-length prefixes in slash notation (such as /22), allowing blocks of any suitable size, efficient address allocation, and route aggregation (supernetting) that reduces the size of routing tables and the load on routers. Give a short example of aggregating several contiguous networks into one prefix. *(5 Marks)*

**(b)**A customer site connects to a service provider's MPLS backbone through its Customer Edge (CE) router, which has the interface address 192.168.25.130 with the subnet mask 255.255.255.192 (/26). The provider edge (PE) router needs the customer's address block in order to associate it with a label-switched path (LSP) or VPN route. Determine, showing the binary working: (i) the Network ID (by ANDing the address with the mask), (ii) the Broadcast Address (by setting all host bits to 1), and (iii) the range of usable host IP addresses (first usable to last usable), along with the block size and the number of hosts in the block. *(5 Marks)*

***(10) (CO 1)(CO2) (BL2)*** 4. An organization is assigned the network address 172.16.0.0/16 and must divide it into at least 6 subnets for different departments while maximizing the number of usable hosts per subnet. Show the working at each step. **(a)** Determine the minimum number of bits to borrow from the host portion (using 2^n ≥ 6) so that the usable hosts are maximized. Write the resulting subnet mask in both dotted-decimal and CIDR notation.*(3 Marks)*

**(b)** Calculate the total number of subnets created (2^n) and the number of usable host IP addresses per subnet (2^h − 2, where h is the number of remaininghost bits). Comment on the number of unused subnets left over. *(3 Marks)*

**(c)** For the first two subnets, list the Network ID, first usable IP, last usable IP, and Broadcast address. Show the block size (increment) used to find the second subnet, and present the results in a table. *(4 Marks)* ***(10) (CO2) (BL3)***

5. Answer the following questions on routing in network simulation tools.

**(a)** Compare and contrast static routing and dynamic routing in network simulation tools such as Cisco Packet Tracer. Compare them on configuration effort (manual versus automatic), adaptability to topology changes and link failures, convergence, scalability, router CPU and memory use, bandwidth overhead of routing updates, security, and administrative distance. Mention dynamic protocols such as RIP, OSPF, and EIGRP, and explain how a simulator lets you observe their behaviour. Discuss the conditions in which static routing is preferred, such as small networks, stub networks with a single exit path, default routes to an ISP, and situations needing predictability, low overhead, or control. Present the comparison in a table. (6 Marks)

**(b)** Outline the basic CLI configuration steps on a router to set up a static route directing traffic for the network 192.168.2.0/24 through the next-hop IPaddress 10.0.0.2. Include thesteps enable, configure terminal, ip route 192.168.2.0 255.255.255.0 10.0.0.2, exit, and saving the configuration with copy running-config startup-config. Then show how to verify the route with show ip route (looking for the S code) and test connectivity using ping . (4 Marks)

***(10) (CO2) (BL3)(BL4)***

SAVEETHA
INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

**Programme: B.E / B.Tech**

**Course Code:** CSA07

**Course Name: Computer Networks Duration: 1: 30 Hr**

**Date : 03**.11.2026 **Max Marks: 50 Marks**

**CONTINUOUS INTERNAL ASSESSMENT 1**

**EVALUATION KEY**

**Q1. OSI Model – Data Link and Transport Layers (CO1, BL2 & BL4) – 10 Marks (a) Data Link and Transport Layer Functions and Reliability – 6 Marks**

| Expected point | Evaluation guideline | Marks |
|---|---|---|
| Data Link layer | Correct description of Data Link layer functions. | 2 |
| Transport layer | Correct description of Transport layer functions. | 2 |
| Collective reliability | Clear explanation of how the two layers together ensure end-to-end and hop-by-hop reliability. | 2 |

**Model answer**

- **Data Link Layer (Layer 2):** Handles node-to-node delivery, framing, physical (MAC) addressing, and error detection over a single physical link.
- **Transport Layer (Layer 4):** Manages end-to-end communication between application processes, segmentation, and end-to-end reliability (acknowledgments, sequence numbers and retransmissions).
- **Collective reliability:** The Data Link layer ensures error-free transmission across each individual hop, while the Transport layer ensures reliable, complete deliveryfrom the original source host to the final destination host across the entire path.

**(b) Connection-Oriented vs. Connectionless Services – 4 Marks**

| Expected point | Evaluation guideline | Marks |
|---|---|---|
| Differentiation | Clear differentiation between connection-oriented and connectionless services. | 2 |
| Examples | Accurate real-world communication examples for both. | 2 |

**Model answer**

- **Connection-Oriented:** Establishes a formal logical connection (handshake) before data transmission. Guarantees delivery and order. Example: **TCP**; placing a telephone call.
- **Connectionless:** Transmits data immediately without prior connection setup. No delivery guarantees at this layer. Example: **UDP**; sending a postal letter.

**Q2. Encapsulation and Network Layer Routing (CO1, BL2) – 10 Marks (a) Encapsulation and Decapsulation – 6 Marks**

| Expected point | Evaluation guideline | Marks |
|---|---|---|
| Encapsulation | Detailed step-by-step description of encapsulation at the source. | 3 |
| Decapsulation | Detailed step-by-step description of decapsulation at the destination. | 3 |

**Model answer**

- **Encapsulation:** As data moves down the OSI layers at the source, each layer adds its own control header (and trailer at the Data Link layer).

| Layer | Unit of data | Added at this layer |
|---|---|---|
| **Application / upper layers** | Data | Application data |
| **Transport** | Segment | Transport header |
| **Network** | Packet | Network header (IP addresses) |
| **Data Link** | Frame | Header and trailer (MAC addresses) |
| **Layer** | **Unit of data** | **Added at this layer** |
| **Physical** | Bits | Signals on the medium |

- **Decapsulation:** As bits are received and moved up the OSI layers at the destination, each layer strips off its corresponding header/trailer, processes it, and passes the payload up to the next layer. **(b) Network Layer Routing and Logical Addressing – 4 Marks**

| Expected point | Evaluation guideline | Marks |
|---|---|---|

| | | |
|---|---|---|
| Layer identified | Correct identification of the Network layer. | 1 |
| Routing | Clear explanation of packet routing across interconnected networks. | 1.5 |
| Logical addressing | Accurate explanation of how logical addressing functions. | 1.5 |

**Model answer**

- **Layer responsible:** Network Layer (Layer 3 – IPv4/IPv6).
- **Logical addressing:** Assigns unique hierarchical logical IP addresses containing network and host portions. Routers inspect the destination IP address in the packet header to make forwarding decisions across interconnected networks.

**Q3. IP Addressing and Subnetting (CO1,CO2, BL2) – 10 Marks (a) Classful Addressing Limitations and CIDR – 6 Marks**

| Expected point | Evaluation guideline | Marks |
|---|---|---|
| Classful limitations | Clear explanation of classful IP limitations (address wastage, scalability issues). | 3 |
| CIDR | Clear explanation of how CIDR/VLSM overcomes these limitations. | 3 |

**Model answer**

- **Classful limitations:** Rigid fixed classes (A, B, C) led to severe address wastage and very large global routing tables because of inflexible block sizes.
- **CIDR:** Removes class boundaries by using flexible prefix lengths (variable masks). This allows address blocks of theright size andsupports route aggregation to keeproutingtables small. **(b) Subnet Calculation – 4 Marks**

| Expected point | Evaluation guideline | Marks |
|---|---|---|
| Network ID | Correct Network ID calculation. | 1.5 |
| Broadcast address | Correct Broadcast Address calculation. | 1.5 |
| Usable range | Correct usable host IP range. | 1 |

**Model answer**

- Given: 192.168.25.130, mask 255.255.255.192 (/26). Block size in the last octet = 256 − 192 = 64. The host address 130 lies in the block 128–191.
- **(i) Network ID:** 192.168.25.128 • **(ii) Broadcast Address:** 192.168.25.191
- **(iii) Usable host range:** 192.168.25.129 to 192.168.25.190 **Q4.**

**Subnetting 172.16.0.0/16 (CO2, BL3) – 10 Marks** Given network 172.16.0.0/16; at least 6 subnets required.

**(a) Subnet Mask – 3 Marks**

| Expected point | Evaluation guideline | Marks |
|---|---|---|
| Bits / CIDR | Correct bit calculation and CIDR notation (/19). | 1.5 |
| Dotted-decimal mask | Correct dotted-decimal subnet mask (255.255.224.0). | 1.5 |

**Model answer**

- Bits borrowed: $2^3 = 8 \geq 6$, so **3 bits** are borrowed. New prefix: **/19**.
- **Dotted-decimal subnet mask:** 255.255.224.0 **(b) Total Subnets and Usable Hosts – 3 Marks**

| Expected point | Evaluation guideline | Marks |
|---|---|---|
| Total subnets | Correct calculation of total subnets. | 1.5 |
| Usable hosts | Correct calculation of usable host IPs per subnet. | 1.5 |

**Model answer**

- **Total subnets:** $2^3 = 8$.

- **Usable hosts per subnet:** host bits = 32 − 19 = 13, so $2^{13}$ − 2 = **8,190**. **(c) First Two Subnets – 4 Marks**

| Expected point | Evaluation guideline | Marks |
|---|---|---|
| Subnet 1 | Accurate Network ID, usable range and broadcast for Subnet 1. | 2 |
| Subnet 2 | Accurate Network ID, usable range and broadcast for Subnet 2. | 2 |

**Model answer**

Block size in the 3rd octet = 256 − 224 = 32.

| Subnet | Network ID | First usable IP | Last usable IP | Broadcast |
|---|---|---|---|---|
| **1** | 172.16.0.0 | 172.16.0.1 | 172.16.31.254 | 172.16.31.255 |
| **2** | 172.16.32.0 | 172.16.32.1 | 172.16.63.254 | 172.16.63.255 |

**Q5. Routing in Network Simulation Tools (CO2, BL3 & BL4) – 10 Marks (a) Static vs. Dynamic Routing and Preferences – 6 Marks**

| Expected point | Evaluation guideline | Marks |
|---|---|---|
| Comparison | Clear comparison/contrast of static and dynamic routing. | 3 |
| Conditions | Accurate identification of operational conditions favouring static routing. | 3 |

**Model answer**

- **Static routing:** Manually set by administrators. Secure and with zero routing-protocol overhead, but does not adapt automatically to failures.
- **Dynamic routing:** Automatically discovers paths using protocols (OSPF, EIGRP). Adapts to topology changes, but consumes CPU and bandwidth.
- **When static is preferred:** Small, stub networks with a single exit point (e.g., an ISP connection), where simplicity, control and zero overhead are required.

**(b) Router CLI Configuration Steps – 4 Marks**

| Expected point | Evaluation guideline | Marks |
|---|---|---|
| Command syntax | Correct mode entry and command syntax for static routing (ip route …). | 2.5 |
| Execution flow | Proper execution flow (global configuration mode, saving/exiting). | 1.5 |

**Model answer**

```
Router> enable
Router# configure terminal
Router(config)# ip route 192.168.2.0 255.255.255.0 10.0.0.2
Router(config)# end
Router# write memory
```

NIKITHAA.B
196526120
computer Network
CSH0709



1) Data link layer:-

The data link layer is responsible for a transfering data between two directly connected devices over a physical Network

functions:-

1) framing : divides data into frames for transmission.

2) Physical addressing : uses MAC addresses to identify devices on a local Network.

3) Error detection : detects transmission errors using methods such as CRC

4) flow control : Helps to prevent a sender from overwhelming a receiver when supported by the link protocol.

2) Transport layer:-

The transport layer provides communication between applications a receiver application running on different devices

* Segmentation :

Divides application data into smaller segments.

* Error Segments : recovery:-

Tcp detects missing data.

| Features | connected - oriented service | connected less service. |
|---|---|---|
| Protocol | TCP | UDP |
| connected setup | Required | Not required |
| Reliability | Support reliable delivery | No delivery guarantee. |
| Acknowledgement | uses acknowledgement | No built in acknowledgement. |
| Data ordering | Maintain ordered delivery | No ordering guarantee. |
| Speed and recovery | generally more overhead. | generally lower overhead. |
| Example | file transfer, web browsing using HTTPS over TCP | online gamingONs queries, live streaming using UDP |

TCP is prefered when reliable ordered delivery is required, whereas UDP is useful when low latency and over overhead are important.

a) Encapsulation :-

It is a process of adding protocol information to data as it moves down the OSI layers at the source device

| OSI Layer | Process | Data Unit |
|---|---|---|
| 7) Application | creates application data | Data |
| 6) Presentation | Handles translation, encryption or the compression as needed | Data |
| 5) Sensor | Manages communication sensors | Data |
| 4) Transport | Adds transport header, such as TCP or UDP | Data / segment |
| 3) Network | Add MAC addressing and from checking information | Packet |
| 2) Data link | Address source | frame |
| 1) Physical | Transmit bits over the medium | Bits |

8) Classfull IP Addressing :-

In classfull addressing, IPv4 addresses were divided into fixed class A, B, C, D and E

| Class | first Octet Range | Default Mask | Purpose |
|---|---|---|---|
| A | 1 - 126 | 255.00(18) | Large Networks |
| B | 128 - 191 | 255.255.00 (116) | Medium Sized |
| C | 192 - 223 | 255.255.0 (285) | Small |
| D | 224 - 239 | Not applicable | Multicast |
| E | 240 - 255 | Not applicable | Reversed. |

CIDR improves IPV4 address utilization and routing efficiency by using flexible network Prefix lengths. Smaller organizations can receive smaller address blocks.

A) Given:

* Network address : 172.16.00/16
* Required Subnets : At least 6

Step 1 : $2^n \geq$ Required Subnets

for 6 Subnets :

$2^2 = 4$

$2^3 = 8$

$\therefore$ 3 bits must be borrowed.

Step 2:

Prefix = /16

Borrowed = 3

New prefix = 16 + 3 = 19

∴ new CIDR prefix is /19 255.255.224.0

∴ CIDR notation: /19

Dottial - Decimal Subnet mask: 255.255.224.0